

04 Assignment - Jenkins

Name: Pooja Prakash Belgaumkar

1. L4 – Automate the CI/CD Pipeline using GitHub webhook and Poll-SCM.

- Create new Freestyle project in the Jenkins.

 **Jenkins** / All ▾ / New Item 🔍 ⚙️ 👤

New Item

Enter an item name

L4-CICD-Pipeline

Select an item type

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 **Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

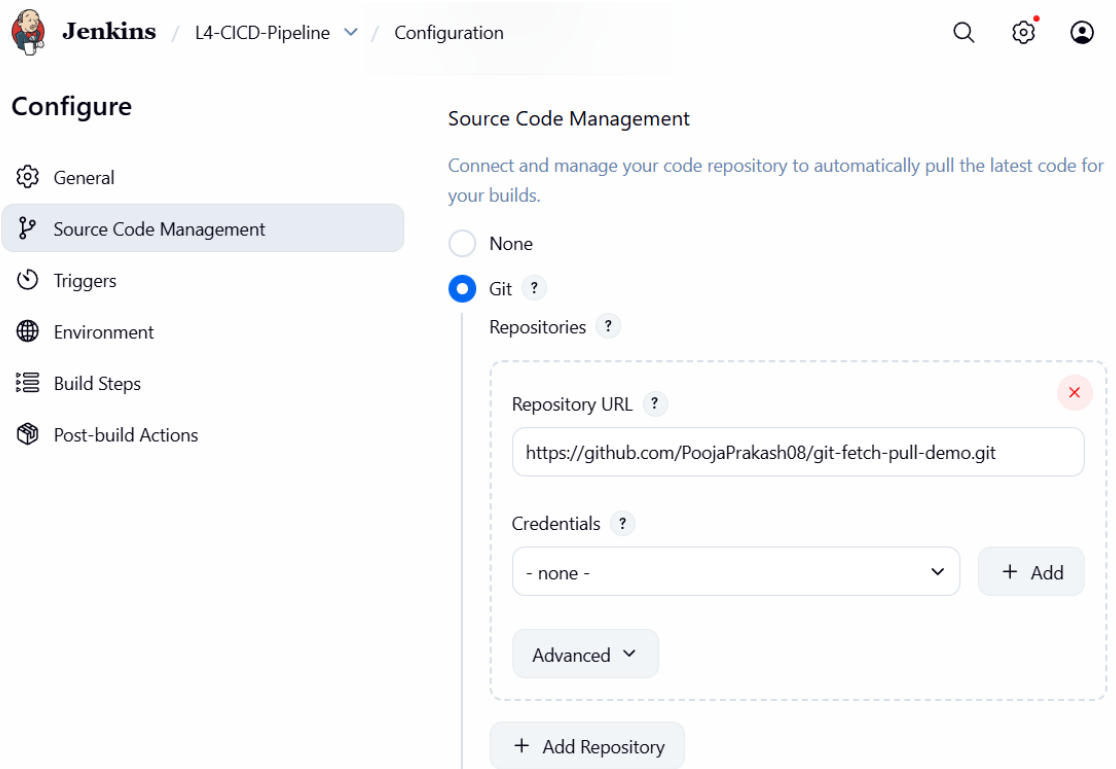
 **Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

 **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 **Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a

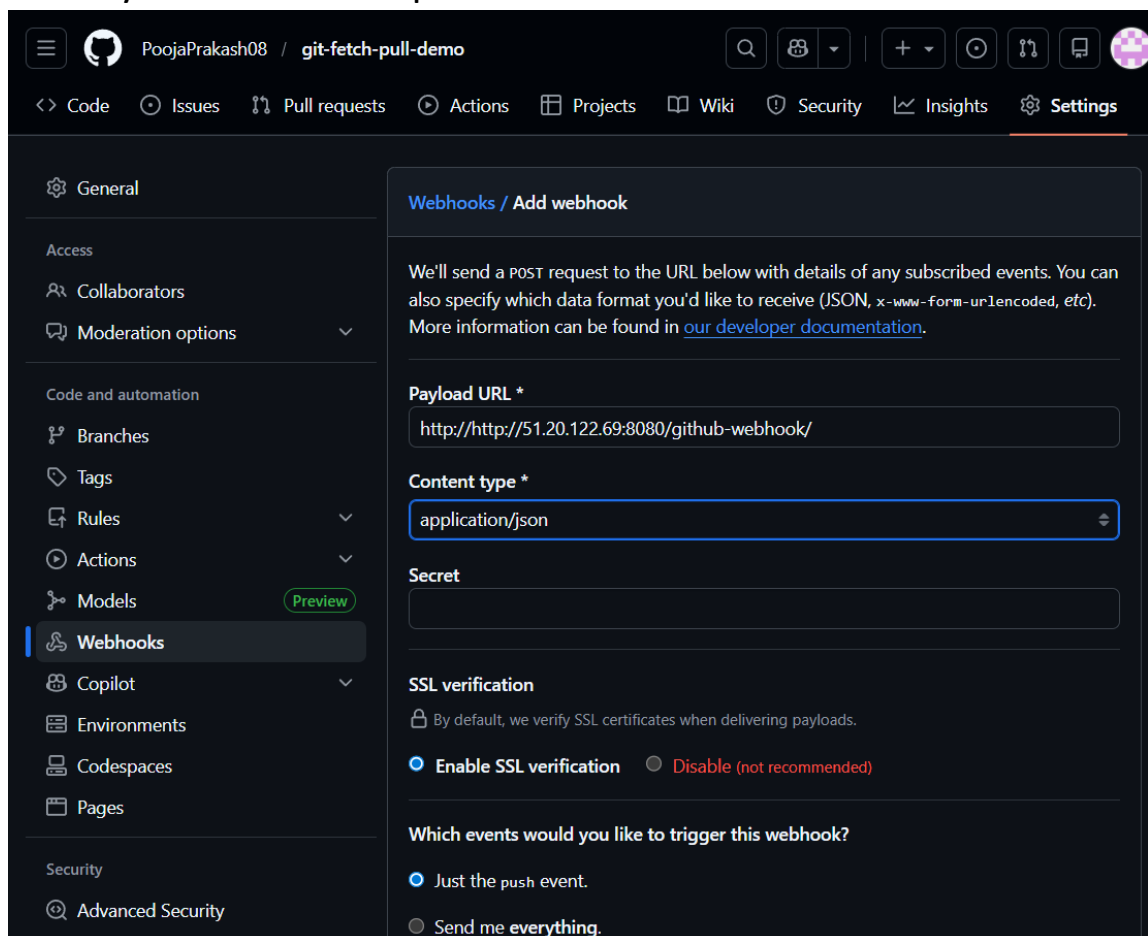
OK

- Enter GitHub repo URL under Source Code Management -> Git.



The image shows the Jenkins Configuration page for Source Code Management. The left sidebar contains a menu with options: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is titled 'Source Code Management' and includes a description: 'Connect and manage your code repository to automatically pull the latest code for your builds.' Below this, there are radio buttons for 'None' and 'Git' (selected). A 'Repositories' section is shown with a dashed border, containing a 'Repository URL' field with the value 'https://github.com/PoojaPrakash08/git-fetch-pull-demo.git', a 'Credentials' dropdown menu set to '- none -', and an 'Advanced' dropdown. A '+ Add Repository' button is at the bottom.

- Then create Webhook in GitHub, this makes GitHub notify Jenkins instantly when there's a push.



The image shows the GitHub Webhook configuration page for the repository 'git-fetch-pull-demo'. The left sidebar contains a menu with options: General, Access, Collaborators, Moderation options, Code and automation, Branches, Tags, Rules, Actions, Models, Webhooks (selected), Copilot, Environments, Codespaces, Pages, Security, and Advanced Security. The main content area is titled 'Webhooks / Add webhook' and includes a description: 'We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in our developer documentation.' Below this, there are fields for 'Payload URL *' with the value 'http://http://51.20.122.69:8080/github-webhook/', 'Content type *' set to 'application/json', and a 'Secret' field. There is also an 'SSL verification' section with a checkbox 'Enable SSL verification' (checked) and a 'Which events would you like to trigger this webhook?' section with radio buttons for 'Just the push event.' (selected) and 'Send me everything.'



Configure



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

[See the list of available environment variables](#)

```
cd maven-webapp
mvn clean install
```



Status



</> Changes



Console Output



Edit Build Information



Delete build '#1'



Polling Log



Timings



Git Build Data



Console Output

Download

Copy

View as plain text

```
Started by an SCM change
Running as SYSTEM
Building on the built-in node in workspace
/var/lib/jenkins/workspace/L4-CICD-Pipeline
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/PoojaPrakash08/git-fetch-pull-demo.git
> /usr/bin/git init /var/lib/jenkins/workspace/L4-CICD-Pipeline # timeout=10
Fetching upstream changes from
https://github.com/PoojaPrakash08/git-fetch-pull-demo.git
> /usr/bin/git --version # timeout=10
```